15. LLM is a type of AI trained on a massive amount of text to understand and generate human-like text.

Difference from standard ML: standard ML might predict a yes / No answer e.g. "Is this email spam?" whereas an LLM can write a 500 word essay about the history of Pretoria.

Example: open AI's GPT-4 (behind chatGPT) or Google's Gemini which are used for complex reasoning tasks.

16. Generative AI refers to AI models that create new content like text, images, audio and code, rather than just analysing existing data.

At a high level, it uses foundational models that have learned patterns of human creativity to predict the next word in a sentence or the next pixel in a photo.

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Three tools: ChatGPT (open AI) for text generation and coding

Midjourney for creating high quality digital art from text prompts

Suno AI for generating full songs and lyrics.

17. Training Dataset is the initial set of data used to teach a ML model how to make predictions or perform a specific task. It is like a textbook that a student uses to study before a final test.

The model looks at thousands or millions of examples in the training set to identify relationships and patterns. For example, to teach AI to recognise a Protea flower, it would need a training dataset with thousands of photos labelled Protea.

The goal is for the model to generalise by learning the patterns well enough to recognise a protea it has never seen before.

Quality is important because a ML model does not have "common sense". It only knows what it is shown. This is referred to Garbage in, Garbage out.

Accuracy: High quality data must be accurate and correctly labelled. If a picture of daisies is

labelled as proteas, the AI will learn the wrong information.

Diversity: Data should represent all possible real world scenarios. For example, a self driving car model needs training data from rainy Cape Town streets, dusty Karoo roads and heavy traffic in Joburg to be safe everywhere.

Completeness: Missing information can lead to gaps in the AI's logic; making it unreliable when it encounters those missing scenarios in the real world.

When the training data is flawed, the AI model can become biased; meaning it unfairly prejudices certain or provides wrong answers.

① If data used to train an AI reflects human prejudices, the AI will automate those prejudices.

Example: If a bank uses AI to approve home loans, but the training data only includes successful loans from wealthy suburbs from 20 years ago, the AI might unfairly reject applications from townships that are still developing; simply because they do not look like historical data. This leads to systemic exclusion because certain groups are denied opportunities like jobs, loans, insurance, based on flawed historical patterns.

② If data is incorrect, the model will make confident

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but wrong predictions.

Example: In healthcare, if an AI is designed to detect skin cancer but the training images only feature light skinned patients, the AI may fail to recognise cancerous cells in darker skin tones. In such cases, life threatening medical errors are caused by a lack of representative data.

③ In LLMs like chatGPT, incorrect training data can lead to hallucinations where the AI makes up facts that sound convincing.

This spreads misinformation, leading people to believe false historical facts or incorrect legal and medical advice.

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18. Computer vision is a field of AI that trains computers to interpret and understand the visual world. Using digital images from cameras and videos, machines can accurately identify and classify objects and

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then react to what they see.

Machines do not see and interpret images the way humans do (as shapes and colours). Instead, they see them as a massive grid of numbers.

① Image as Data: computer perceives image as a matrix of pixels. Each pixel has a numerical value representing its colour and brightness.

② Preprocessing: System cleans the image by resizing it or removing blurriness to make numbers easier to process.

③ Feature Extraction: using convolutional Neural Networks (CNNs), the AI scans the image for patterns.

First layer → detect simple things like edges or lines

Middle layers → combine those lines to find shapes like a circle or square.

Final layers → combine shapes to identify complex objects like face or car.

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④ Classification → system compares these patterns against its training dataset. If the patterns match its memory of a car, it labels the image as car with a certain percentage of confidence.

Examples

① An SA bank uses computer vision to automate the monitoring of the cash replenishment process at its ATMs.

cameras at the ATM use Human Activity Recognition to detect exactly what security guards and technicians are doing. Ensures that safety protocols are followed and can istantly flag suspicious behaviour or deviations from standard procedure; significantly reducing risk of cash in transit heists or internal fraud.

② Agritech companies like Aerobotics and Zindi use computer vision to help farmers manage their crops

Drones or satellites capture high resolution images of orchards a fields. The computer vision model analyses the images to identify individual trees and detect early signs of pest infestations or diseases invisible to the naked eye.

Allows farmers to treat only the affected trees instead of spraying a whole farm with chemicals; saving money & protecting environment.

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